

Vicky Kumar

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Summary

Software Engineer who builds real backend systems and AI applications. Experienced in **Python and FastAPI**, with hands-on work in **real-time APIs, WebSockets, and AI-driven systems**. Have built, deployed, and debugged **production-ready projects** end to end. Comfortable taking ownership from design to deployment.

Experience

Mercor, Outlier, Soul AI

2024 - 2025

AI Evaluation & Prompt Engineer (Contract)

Remote

- Evaluated **thousands of LLM outputs** using structured rubrics (accuracy, reasoning, safety, instruction-following)
- Identified hallucination patterns, edge cases, and multi-turn failure modes in agent workflows
- Designed prompt improvements and feedback loops to improve **response reliability in production models**
- Tested tool-using agents and ranked outputs for reinforcement-style feedback pipelines

Projects

AI Bid Writer — LLM-Based Agent System | <https://freelancer-com-omega.vercel.app/>

- Trained CNN models on a 4,514-image MRI dataset with augmentation (Elastic, CLAHE, Noise)
- Achieved **97.94% accuracy** and **98.69% precision** using staged fine-tuning
- Built full-stack deployment with React UI, FastAPI inference server, and CI/CD on Render
- Created real-time training dashboards and mobile-friendly prediction interface
- EfficientNetB3 | TensorFlow | FastAPI | React | Docker**

Brain Tumor Detection System | <https://brain-tumor-mcug.onrender.com/>

- Designed and trained a CNN-based model on **~4,500 MRI scans**, achieving **97.9% accuracy** and **98.7% precision**
- Built a complete inference pipeline: image upload → preprocessing → prediction → result visualization
- Deployed the model using **FastAPI** with containerized inference via **Docker** and automated CI/CD
- Developed a responsive frontend to enable real-time clinical-style interaction with the model
- Python, PyTorch, EfficientNet, FastAPI, React, Docker, CI/CD**

QuickPoll — Real-Time Distributed Voting Platform | <https://quickpoll-frontend-7nyx.onrender.com/>

- Architected a real-time voting system supporting **concurrent users** with **sub-50ms latency**
- Implemented WebSocket-based communication to enable live vote updates without polling
- Enforced vote integrity using backend validation and database-level constraints
- FastAPI (async), WebSockets, PostgreSQL, Docker**

Edited.Frame – Professional Video Editor Portfolio | <https://algsoch.github.io/sachin>

- Built a modern, responsive video portfolio optimized for **9:16 aspect ratio content** with a dynamic grid layout (mobile → desktop).
- Integrated a **Firestore-powered real-time review system** with star ratings, approval workflow, and satisfaction metrics.
- Designed a **glassmorphism-based UI** with responsive breakpoints, premium typography, and smooth animations.
- Featured **advanced video categorization** (trading, educational, motion graphics, social media edits) with a lightbox player and filtering.

TDS Tool-Based Assistant | <https://assistant-chatbot-izrt.onrender.com/>

- Built a backend assistant capable of routing user queries to **50+ internal tools**
- Implemented tool-selection logic for file processing, parsing, scraping, and automation tasks
- Exposed functionality via clean REST APIs with structured outputs
- Python, FastAPI, LangChain, HTML/CSS/JS**

Skills

- Languages:** Python, SQL, JavaScript, Java
- Backend & APIs:** FastAPI (async), REST APIs, WebSockets, API Design, Authentication & Authorization
- Databases:** PostgreSQL, MySQL, MongoDB
- AI / ML:** PyTorch, CNN-based Models, Model Training & Evaluation, Inference Pipelines
- LLMs & Agentic Systems:** LLM APIs (Gemini, OpenAI), Prompt Engineering, LangChain, LangGraph, CrewAI, Tool-Calling Agents, Multi-Agent Workflows, RAG
- Frontend:** React, Tailwind CSS, HTML5, CSS3
- DevOps & Engineering Tools:** Docker, CI/CD Pipelines, Git, GitHub Actions

Education

IIT Madras

May 2024 - May 2028

B.S., Data Science and Applications

Chennai, Tamil Nadu

- Coursework:** Python Foundation, Mathematics & Statistics, Machine Learning Foundation and Technique, Data Analytics, Tool Of Data Science

Certifications & Achievements

- **Participant:**The Unstoppable Hackathon (Innovation Track), LNMIIT Jaipur (Dec 2025)
- **Participant:**Model Masters, InnoVision'25, Netaji Subhas University of Technology (Nov 2025)
- **WorkShop:**NLP with Transformers — IIT Madras (2025)
- **Foundational Completion:**Foundational Level in Programming & Data Science — IIT Madras (2024)